User Testing Protocol: CHD Bayesian Predictor Prototype

Date: [DD/MM/YYYY]

Name: [Tester Name]

Tester Background: [e.g., Medical Student, CS Peer, General User]

1. Introduction & Briefing

"Thank you for testing our prototype. This application uses a Bayesian network trained on a dataset of 300 patients to calculate the probability that a patient has Coronary Heart Disease (CHD) based on their symptoms.

Important: We are testing the usability of the software, not your medical knowledge. You do not need to be a doctor to do this. Please 'think aloud' as you use the site and let us know if anything is confusing, slow, or broken."

2. Pre-Test Questionnaire

To understand the tester's baseline perspective.

1. What is your general background? (e.g., Tech, Healthcare, Student) Answer: [Blank space]

2. What is your general comfort level with health or medical apps? Scale 1 to 5 (1 = Very Uncomfortable, 5 = Very Comfortable): [Blank space]

3. Have you ever used a medical symptom checker or health tracking app before? Answer (Yes, No, Not sure): [Blank space]

4. How would you rate your existing knowledge of Coronary Heart Disease (CHD)? Scale 1 to 5 (1 = No prior knowledge, 5 = Highly knowledgeable): [Blank space]

5. On a scale of 1 to 5, how comfortable are you interpreting statistical probabilities like percentages? Scale 1 to 5 (1 = Not comfortable, 5 = Very comfortable): [Blank space]

3. Testing Tasks

Provide the tester with the following "Mock Patient Data" to use during the test.

Mock Patient Profile A: Age/Sex: 58, Male Vitals: Blood Pressure 150/95, Heart Rate 88 bpm Labs: Total Cholesterol 240 mg/dL Symptoms: Chest tightness, Smoker (Yes)

Task 1: First Impressions Goal: Assess initial clarity and trust. Prompt: "Without entering any data yet, what is your first impression of this page? Is it clear what you are supposed to do?" Observations / Feedback: [Blank space]

Task 2: Data Entry (Usability & Speed) Goal: Evaluate the form UI, input validation, and layout. Prompt: "Please enter the data for Mock Patient Profile A into the system to get a risk estimate." Time taken: [Mins : Secs] Observations (Did they struggle with formatting or mapping the mock data to the dropdown categories? Were dropdowns or text fields confusing? Did terms like 'Angina' cause hesitation?): [Blank space]

Task 3: Output Interpretation Goal: Ensure the Bayesian calculation is presented clearly. Prompt: "Look at the percentage result. What does this number mean to you? Is it clear how the system arrived at this conclusion? How do you feel about the way the AI Doctor explained the results on the right side of the screen?" Observations (Does the user trust the result? Do they want more explanation?): [Blank space]

Task 4: Using the "What-If" Feature Goal: Test discovery of interactive AI scenarios. Prompt: "Let's say you want to know how starting a Statin therapy would change this patient's risk. How would you figure that out using this tool?" Observations (Did they use the AI Chat box or change the dropdown on the left?): [Blank space]

Task 5: Understanding the Breakdown Goal: Evaluate comprehension of the mathematical and clinical metrics. Prompt: "Please scroll down to the Risk Factor Breakdown and Model Clinical Verification sections. Was it clear to you which factors increased the risk and which lowered it? Do you understand what the verification section is trying to tell you about the AI's accuracy?" Observations (Do they understand the green/red bars and terms like Sensitivity/Specificity?): [Blank space]

Task 6: Missing Data / Error Handling Goal: Test the robustness of the Bayesian network and form logic. Prompt: "Now, let's pretend we don't have the patient's cholesterol data. Try submitting the form with that field completely blank." Observations (Did the system crash? Did it update the probability successfully using partial data?): [Blank space]

4. Post-Test Ratings

Prompt: "Please rate your agreement with the following statements based on your experience today." (Use a 1 to 5 scale: 1 = Strongly Disagree, 5 = Strongly Agree)

1. The tool was easy to navigate.

2. The medical terminology was easy to understand.

3. The visual design of the interface is appealing and professional.

4. The AI Doctor’s explanation helped me understand the score better.

5. The difference between the mathematical Bayesian Network score and the Interactive AI Doctor chat was clear to me.

6. I understand how specific risk factors affected the overall score.

7. The way the Risk Factor Breakdown displayed positive (red) and negative (green) impacts made logical sense.

8. I would feel confident using this tool to learn about how different health factors impact heart disease.

9. Overall, I found the information presented by the calculator to be highly useful.

10. The results generated by this tool felt accurate and reliable based on the data I entered.

11. I would trust the results of this calculator for educational purposes.

5. Post-Test Questionnaire (Open-Ended)

To gather overall impressions.

1. What was the easiest part of using this tool? Answer: [Blank space]

2. What was the most confusing or frustrating part? Answer: [Blank space]

3. What was your favorite feature? Answer: [Blank space]

4. If you were a doctor using this in a rush, what is one thing you would change about the layout or design? Answer: [Blank space]

5. Did you feel you could trust the prediction result? Why or why not? Answer: [Blank space]

6. Overall, do you agree with the final outcome and layout of the interface? Did it feel complete, or is something missing? Answer: [Blank space]

7. In your own words, do you feel the tool we created is both useful and accurate? Why or why not? Answer: [Blank space]

8. Where do you think this application could be most useful in the real world? (e.g., in a classroom, at a doctor's office, for personal use at home) Answer: [Blank space]